

My Study Notes

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1 Introduction

Welcome to my study notes. This document is organized into multiple chapters to keep everything clean and modular.

2 Conjectures and Theorems

2.1 Basic Definitions

Definition 1. A conjecture is a mathematical statement that is believed to be true based on evidence or intuition, but which has not yet been proved or disproved.

Definition 2. A theorem is a mathematical statement that has been rigorously proved to be true from previously established results (such as axioms, definitions, and other theorems).

2.2 Examples from Number Theory

Example 2.1 (A true conjecture that became a theorem). Historically, mathematicians conjectured that there are infinitely many prime numbers. This was later proved by Euclid and is now a theorem:

Theorem 1 (Euclid). There are infinitely many prime numbers.

Thus, a statement can start as a conjecture and, once proved, become a theorem.

Example 2.2 (A conjecture that is still open). Goldbach's Conjecture is one of the most famous unsolved problems in number theory:

Conjecture 1 (Goldbach). Every even integer $n \geq 4$ can be written as the sum of two prime numbers.

This statement has been checked for very large values of n , but no complete proof or counterexample is known.

Example 2.3 (A false conjecture). Consider the following statement:

Conjecture 2. Every odd integer $n > 1$ is a prime number.

At first glance, small examples like 3, 5, 7 might suggest this is true. However, it is false. For instance,

$$9 = 3 \cdot 3, \quad 15 = 3 \cdot 5,$$

so 9 and 15 are odd but not prime. These are counterexamples that disprove the conjecture.

2.3 Remarks

Remark 1. *Conjectures play a central role in mathematical research. They guide exploration and often lead to new techniques, whether they eventually become theorems or are disproved by counterexamples.*